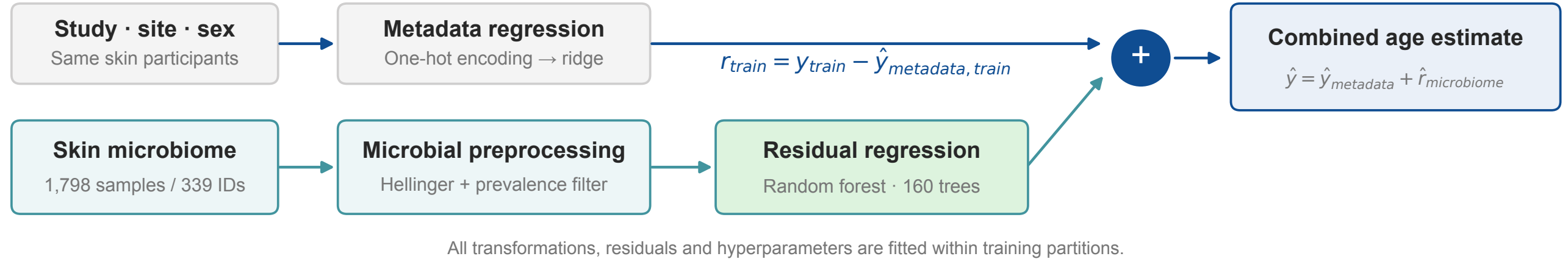


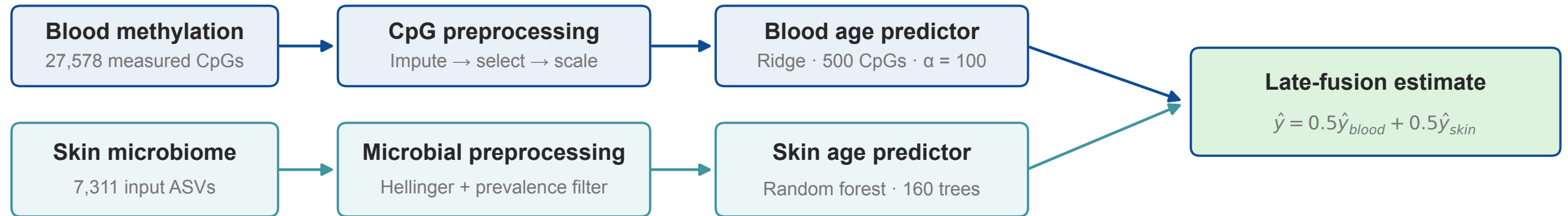
a Metadata-adjusted microbial age prediction

Evaluated using three repeated nested participant-held-out validations



b Blood–skin prediction-level fusion

Fitted component models; joint accuracy requires paired test specimens



Verified shared participant / visit IDs required at inference · equal weights specified without paired calibration.